

Def.

Let $f: [a, b] \rightarrow \mathbb{C}$ be a function. Given a partition $P = \{x_0, \dots, x_n\}$, define

$$V(f, P) \stackrel{\text{def}}{=} \sum_{i=1}^n |f(x_i) - f(x_{i-1})|$$

which is called the variation of f with respect to a partition P . And, the value

$$TV(f) \stackrel{\text{def}}{=} \sup \{V(f, P) \mid P \text{ is partition on } [a, b]\}$$

is called the total variation of f

If the total variation is finite, we say f is of bounded variation.

Prop.

If $f: [a, b] \rightarrow \mathbb{C}$ is continuously differentiable, then f is of bounded variation with

$$TV(f) = \int_a^b |f'(t)| dt$$

Proof. Given a partition $P = \{x_0, \dots, x_n\}$ on $[a, b]$, by the fundamental theorem of calculus,

$$V(f, P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \sum_{i=1}^n \left| \int_{x_{i-1}}^{x_i} f'(t) dt \right| \leq \sum_{i=1}^n \int_{x_{i-1}}^{x_i} |f'(t)| dt = \int_a^b |f'(t)| dt$$

Since P was arbitrary, $TV(f) \leq \int_a^b |f'(t)| dt$, so the total variation has finite value.

To show that the reverse inequality, we use the fact that:

f' is continuous uniformly, because it is continuous on the compact set $[a, b]$.

Given $\varepsilon > 0$, take $\delta_1 > 0$ such that $|x - t| < \delta_1 \Rightarrow |f'(x) - f'(t)| < \varepsilon$.

And, take $\delta_2 > 0$ such that if a partition P on $[a, b]$ satisfies $\|P\| < \delta_2$, then

$$\left| \int_a^b |f'(t)| dt - \sum_{i=1}^n |f'(z_i)| \cdot |x_{i+1} - x_i| \right| < \varepsilon$$

where $P = \{x_0, \dots, x_n\}$ and $z_i \in [x_i, x_{i+1}]$ is arbitrary. The existence of δ_2 is given by the fact that $|f'|$ is Riemann-integrable.

Prop. Let $f: [a, b] \rightarrow \mathbb{R}$ be a function.
Given $c \in (a, b)$,

$$TV_a^b(f) = TV_a^c(f) + TV_c^b(f)$$

Proof. Given partitions P on $[a, c]$ and Q on $[c, b]$, the union $P \cup Q$ is a partition on $[a, b]$. Therefore, the equality given by definition

$$V_a^c(f, P) + V_c^b(f, Q) = V_a^b(f, P \cup Q) \leq TV_a^b(f)$$

holds. Since P and Q were arbitrary,

$$TV_a^c(f) + TV_c^b(f) \leq TV_a^b(f).$$

To obtain the reverse inequality, let R be a partition on $[a, b]$.
Then,

$$\begin{aligned} V_a^b(f, R) &\leq V_a^b(f, R \cup \{c\}) = V_a^c(f, (R \cup \{c\}) \cap [a, c]) + V_c^b(f, (R \cup \{c\}) \cap [c, b]) \\ &\leq TV_a^c(f) + TV_c^b(f). \end{aligned}$$

Since R was arbitrary,

$$TV_a^b(f) \leq TV_a^c(f) + TV_c^b(f)$$

So, Proved. ~~Q~~

Directly, $V(x) = TV_a^x(f)$ is well-defined monotone increasing function, if $f \in BV$.

Prop. Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded variation.
 Then, for any $x \in [a, b)$, the right-sided limit $f(x^+)$ exists.
 Similarly, for any $x \in (a, b]$, the left-sided limit $f(x^-)$ exists.
 Proof. Since f is of bounded variation, a map

$$V: [a, b] \rightarrow \mathbb{R} : t \mapsto TV_a^t(f)$$

is well-defined. Let $x \in [a, b)$ be given. Then, given $x < t_1 < t_2 \leq b$,

$$|f(t_2) - f(t_1)| \leq TV_{t_1}^{t_2}(f) = V(t_2) - V(t_1)$$

Let $\{x_n\}$ be a sequence in $[x, b]$ converging to x .
 Since V is monotone increasing, $V(x^+) = \inf\{V(t) \mid t > x\}$ exists,
 therefore a sequence $\{V(x_n)\}$ converges to $V(x^+)$.

Now, given $\varepsilon > 0$, there are $N_1, N_2 \in \mathbb{N}$ such that
 $n \geq N_1 \Rightarrow |V(x_n) - V(x^+)| < \frac{\varepsilon}{2}$ and $m \geq N_2 \Rightarrow |V(x_m) - V(x^+)| < \frac{\varepsilon}{2}$
 so $m, n \geq \max\{N_1, N_2\}$ implies

$$|V(x_n) - V(x_m)| \leq |V(x_n) - V(x^+)| + |V(x^+) - V(x_m)| < \varepsilon$$

and since $|f(x_n) - f(x_m)| \leq |V(x_n) - V(x_m)|$, $\{f(x_n)\}$ is also a Cauchy sequence,
 so the limit $f(x^+)$ of $\{f(x_n)\}$ exists. Since the sequence $\{x_n\}$ in $(x, b]$ was
 arbitrary, the right-sided limit $f(x^+)$ exists.

The left-sided limit exists by the same way. $\square$

Note: But, the Converse is not true, let

$$f: [0, 1] \rightarrow \mathbb{R} : \begin{cases} 0 & \text{if } x=0 \\ x \cdot \sin \frac{1}{x} & \text{if } x \neq 0. \end{cases}$$

Then, f is not bounded variation, because: Consider the Partition on $[0, 1]$

$$P_n = \left\{ 0, 1, \frac{2}{\pi}, \frac{2}{3\pi}, \dots, \frac{2}{(2n+1)\pi} \right\}$$

The variation $V(f, P_n) \geq \frac{2}{\pi} \cdot \sum_{k=0}^{n-1} \frac{1}{2k+1}$ so, $V(f, P_n) \rightarrow \infty$ as $n \rightarrow \infty$. But,

$$|f(x)| \leq \left| x \cdot \sin \frac{1}{x} \right| \leq |x| \rightarrow 0 \text{ as } x \rightarrow 0^+,$$

so the right-side limit exists.